Daniyal Khan

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PROFILE

Data Science student with 1.5+ years of experience in project work. Equipped with a solid foundation in machine learning and deep learning algorithms and their practical applications. Skilled in implementing end-to-end model pipelines, optimizing models, and deploying solutions through MLOps practices.

EDUCATION

- Masters in Computer Science and Engineering
  Jamia Millia Islamia (2022-24)
  9.29 SGPA
- Bachelors in Computer Science and Engineering
  Galgotias University (2017-21)
  8.16 CGPA
- Intermediate
  Dayawati Modi Academy (2016-17)
  90.33%

CERTIFICATES

- Advanced Learning Algorithms offered by Deeplearning.AI and Stanford on Coursera
- Supervised Machine Learning: Regression and Classification offered by Deeplearning.AI and Stanford on Coursera
- Deeplearning Specialization offered by Deeplearning.AI and Stanford on Coursera (in progress)

ACHIEVEMENTS

- Gate Qualified (97.33 percentile)
- 2nd position in technical quiz conducted by Google Developer Students Club

EXTRACURRICULAR

- Active participant in webinars and workshops related to Machine Learning
- Member of WORKForCompassion NGO, working to promote peace and spread Compassion in the society.

SKILLS

- **Machine Learning**
  Linear Regression, Logistic Regression, Naive Bayes, KNN, SVM, Decision Trees, Random Forest, Boosting (AdaBoost, Gradient Boosting, XGBoost) and Bagging Techniques, PCA, T-SNE, GMM, K-means, Hierarchical clustering, DBSCAN, HMM
- **Deep Learning**
  ANN, CNN, RNN, Transfer learning, Seq2Seq models (Encoder Decoder, Transformer, LLM), Auto-encoders, GANs
- **Statistics**
  Descriptive stats, Inferential stats, Hypothesis testing, EDA, Regression Analysis
- **Techniques**
  Regularization, Cross-Validation, Encoding, Hyper-parameter Optimization, Feature Selection and Construction, Imputation techniques, Pipelines, Handling Imbalance and Multicollinearity
- **MLOPs**
  DVC, DVClive, MLflow, CI/CD, git, github, Containerization (Docker), fastAPI, Pydantic, Flask, Cloud (ECR, EC2, S3), Distributed Computing(Kubernetes), Seldon, prometheus, Kubeflow pipelines, Postman, AutoML
- **Languages**
  Python, SQL
- **Python Packages and frameworks**
  Numpy, Pandas, Matplotlib, Seaborn, SciPy Tensorflow, Scikit-Learn, BeautifulSoap

PROJECTS

Diabetic Retinopathy Diagnosis using CNN with Interpretability

- Preprocessing like resizing, normalisation done, CLAHE and Gaussian filter applied, did Augmentation
- Transfer learning (InceptionV3) used, did hyper-parameter tuning and got SOTA kappa score 0.82
- Employed Grad-CAM and LIME techniques for explaining predictions

Movie recommendation system

- Data collected using web scrapping and imdb api, Data cleaning and data formatting done.
- Combined content filtering and collaborative filtering for better recommendations used.
- Tracked using DVC, MLflow and git, app deployed on AWS EC2.

Multimodal Sentiment Analysis

- Audio and text modalities used. Used pre-trained embeddings for text and melSpectrogram technique for audio.
- Transformer used for encoder (extract features from audio, text) and decoder (tensor fusion)